

Aryan Prakash

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Education

University of California, San Diego

Bachelor of Science in Bioinformatics, Minor in Finance

San Diego, CA

Sep 2023 – Mar 2027

Relevant Coursework: Advanced Data Structures, Design and Analysis of Algorithms, Biological Databases, Applied Genomic Technologies, Accounting, Enterprise Finance, Molecular Sequence Analysis, Metabolic Biochemistry, Genome Diversity and Dynamics, Immunology, Introduction to Data Science, Statistics, Making of the Modern World

Skills

- **Programming:** Python, SQL, R, Java, C++
- **Web Development:** React, Next.js, Supabase (PostgreSQL), REST API, Git, Vercel
- **Bioinformatics and Data Analysis:** RNA-seq analysis, STAR, DESeq2, BLAST, Biopython, PCA, FoldX, PyMOL, Pathway Enrichment, Batch Effect Correction, k-means clustering, Genome Assembly through de Bruijn Graphs, Sequence Alignment
- **Other:** RAG, OpenCV, Financial Modeling, DCF Valuation, MS Office, AI Assisted Development (Cursor, ClaudeCode)

Experience

Bioinformatics Researcher — Cherqui Lab, UCSD Health

June 2025 – Feb 2026

- Analyzed proteomics and RNA-seq datasets for the Friedreich's Ataxia gene therapy project using Python based statistical workflows, including QC, PCA, differential expression analysis, and bioinformatics tools such as STAR and DESeq2.
- Synthesized and parsed complex biological data into actionable findings, answering biological questions and communicating methods and results to scientists to guide future therapeutic target selection.

Bioinformatics Intern — Wild Genomics

May 2025 – Aug 2025

- Collaborated on developing an end-to-end environmental-DNA metagenomics pipeline, transforming raw sequencing data from eDNA samples into interpretable taxonomic outputs through preprocessing and demultiplexing, dereplication, chimera filtering, clustering and singleton removal, and species identification.
- Packaged pipeline outputs into standardized client-ready reports (TSV) for agricultural customers, summarizing taxonomic profiles and species abundance across samples.

Project Member — Wild Genomics (Remote)

Jun 2024 – Oct 2024

- Collaborated with scientific and business teams to conduct market and competitive research and build nuanced pro-forma financial statements to support operational decision-making and strategy.

Intern - VC Grid

Jun 2024 – Sep 2024

- As part of the sourcing team, screened 20+ pitches, rigorously evaluated product-market fit and founder capabilities to filter high-potential investment opportunities for the Investment Committee.
- Built and worked on DCF and financial models to identify valuation gaps, supporting crucial decision-making.

Publications & Projects

RagaRadar

Feb 2026 - Present

Founder

- Built and deployed ragaradar.com, a full-stack web-based interface for discovering Indian Classical Music concerts across North America, creating UI design and database-backed workflows for collecting, organizing, and displaying event data. Over 25 events live on the site in the first 30 days with events ranging across 8 states.

AI Parkinson's Diagnostic Tool

Oct 2025

Winner — Healthlink Hackathon, Track 1, UC San Diego

- Built an early Parkinson's screening prototype that analyzed keystrokes, voice, and eye blinks using LLMs, OpenCV, and Fourier transforms to generate patient risk reports for accessible early screening along with a RAG based chatbot.

Publications: *Treating B-Cell ALL with CAR NK Cells (Horizon Academic Research Journal, 2022); Investigating Stability of a Mutant REG3A Protein using FoldX and PIC (IJSR, 2022)*

Additional

Founder & President — Indian Fusion Music Club - "Khoj"

May 2024 – Jun 2025

- Founded an Indian fusion music club, with 10+ members actively performing at various venues across San Diego

Performing Musician

Jan 2024 – Present

- Professional harmonium player (Indian classical music instrument); performed across Southern California at multiple venues including the prestigious Music Circle and the Festival of Tabla.